

# HybPiper Installation Guide for macOS (Apple Silicon)

## Prerequisites

- Apple Silicon Mac (M1, M2, M3 or newer)
  - macOS Ventura or newer
  - Internet connection
  - Administrator rights to install software
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## 1. Install Miniconda

Download and install the latest **Miniconda for Apple Silicon (macOS ARM64)**:

<https://www.anaconda.com/download/success>

Choose:

**Miniconda** → **macOS** → **Apple Silicon (.pkg)**

Complete the installation using the default settings.

Open a **new Terminal** after the installation has finished.

Not sure if it's always needed, but restarting your laptop can be handy at this point, so that the necessary changes take effect.

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## 2. Configure Conda

Run the following commands:

```
conda config --add channels conda-forge
conda config --add channels bioconda
conda config --set channel_priority strict
```

Verify the channel order:

```
conda config --show channels
```

The output should be:

```
channels:
- bioconda
```

- conda-forge
  - defaults
- 

### 3. Create the HybPiper environment

Create a dedicated Conda environment and install HybPiper:

```
conda create -n hybpiper python=3.12 hybpiper
```

When prompted:

Proceed ([y]/n)?

Type:

y

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### 4. Activate the environment

```
conda activate hybpiper
```

Your prompt should now begin with:

(hybpiper)

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### 5. Verify the installation

Run the following commands:

```
hybpiper --version  
hybpiper check_dependencies
```

The dependency check should report that all required external tools are found, including:

- BWA
- BLAST+
- Exonerate
- GNU Parallel
- SPAdes
- Samtools
- BMap
- DIAMOND

- MAFFT
- 

## 6. Daily usage

Whenever you want to run HybPiper:

Activate the environment:

```
conda activate hybpiper
```

Run your analyses as usual.

When finished:

```
conda deactivate
```

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## 7. Updating HybPiper

To update HybPiper and all dependencies:

```
conda activate hybpiper  
conda update --all
```

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## 8. Removing the environment (optional)

If you ever want to completely remove HybPiper:

```
conda env remove -n hybpiper
```

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## Notes

- This installation has been successfully tested on an Apple Silicon Mac using native ARM64 packages.
- No Rosetta installation is required.
- No manual installation of BWA, SPAdes, Samtools, BLAST or other dependencies is necessary.
- All required software is installed automatically through Bioconda.